



1. MOTIVATION / PROBLEM

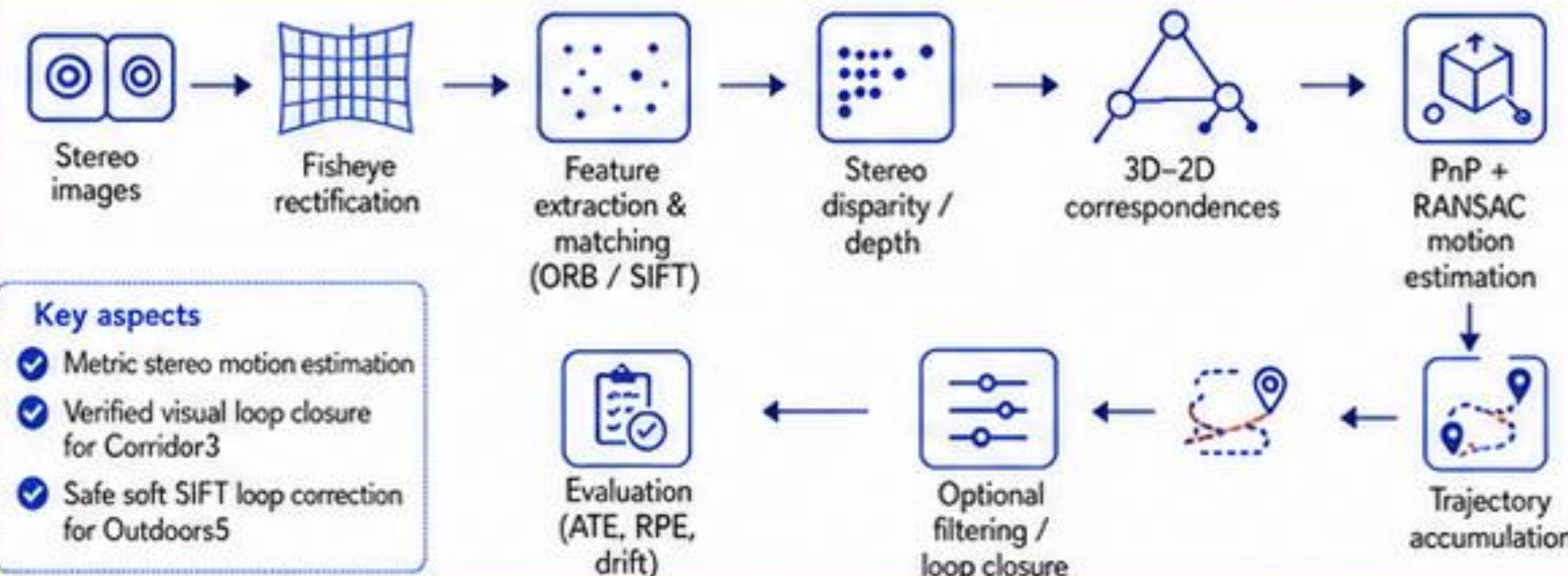
- Stereo VO estimates camera motion from synchronized left-right image pairs.
- Goal:** evaluate a classical geometry-based stereo VO pipeline on TUM VI sequences.
- Focus:** drift behavior, sequence-specific evaluation, and the effect of visual loop closure.
- Constraints:** classical computer vision only. no deep learning, official calibration preserved.

2. DATASETS & EVALUATION PROTOCOL

Sequence	Frames	GT Availability	Evaluation Metric
Room2	2882	Full ground truth available	ATE and RPE
Corridor3	5802	GT available only at start/end	Start-end drift
Outdoors5	17747	Sparse GT coverage (start/end)	Start-end drift

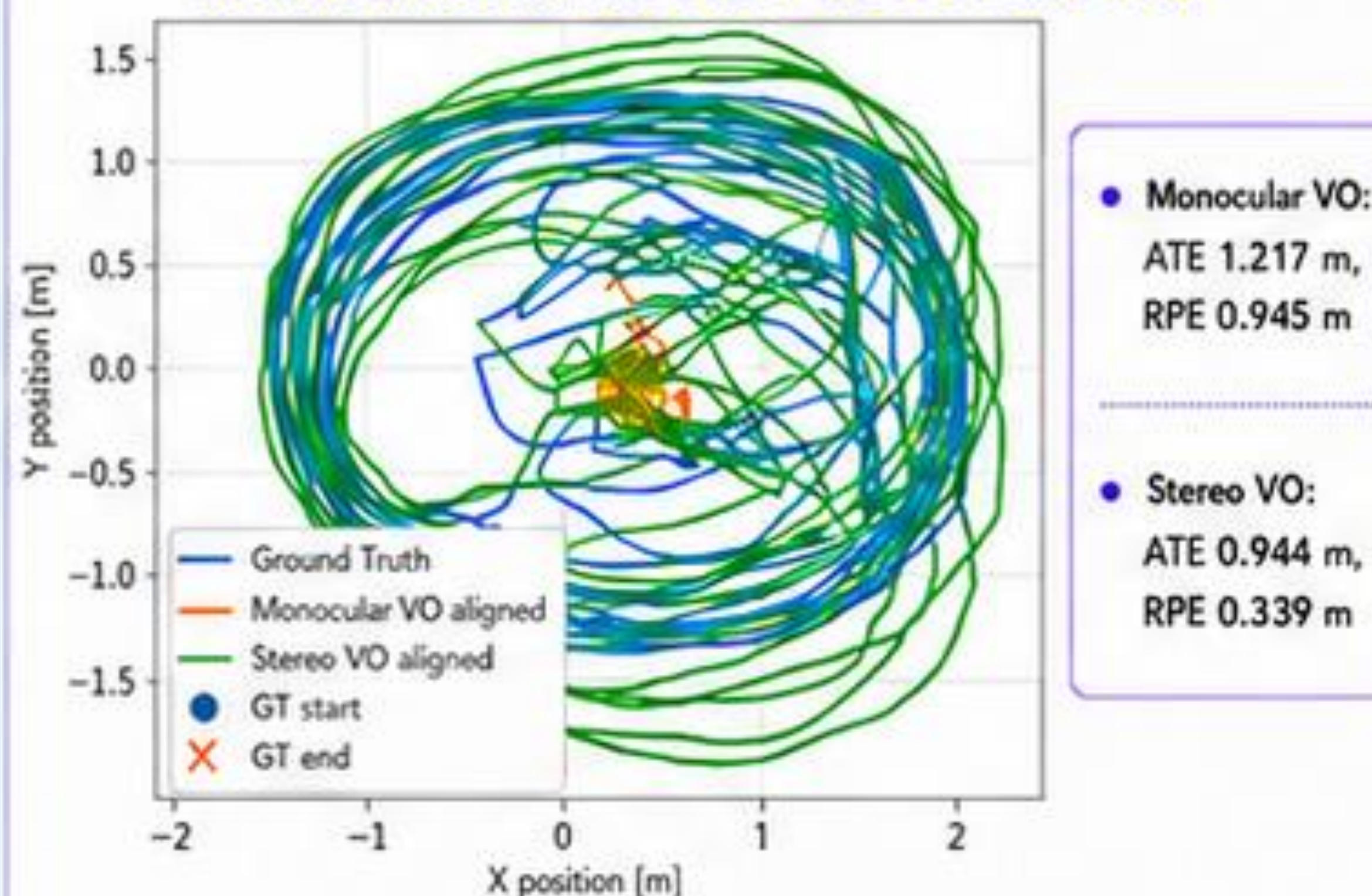
- Calibration:** official TUM VI / Kalibr camchain.
- Important:** do not modify official frame ranges or calibration. Preserve the evaluation protocol.

3. METHODOLOGY / PIPELINE

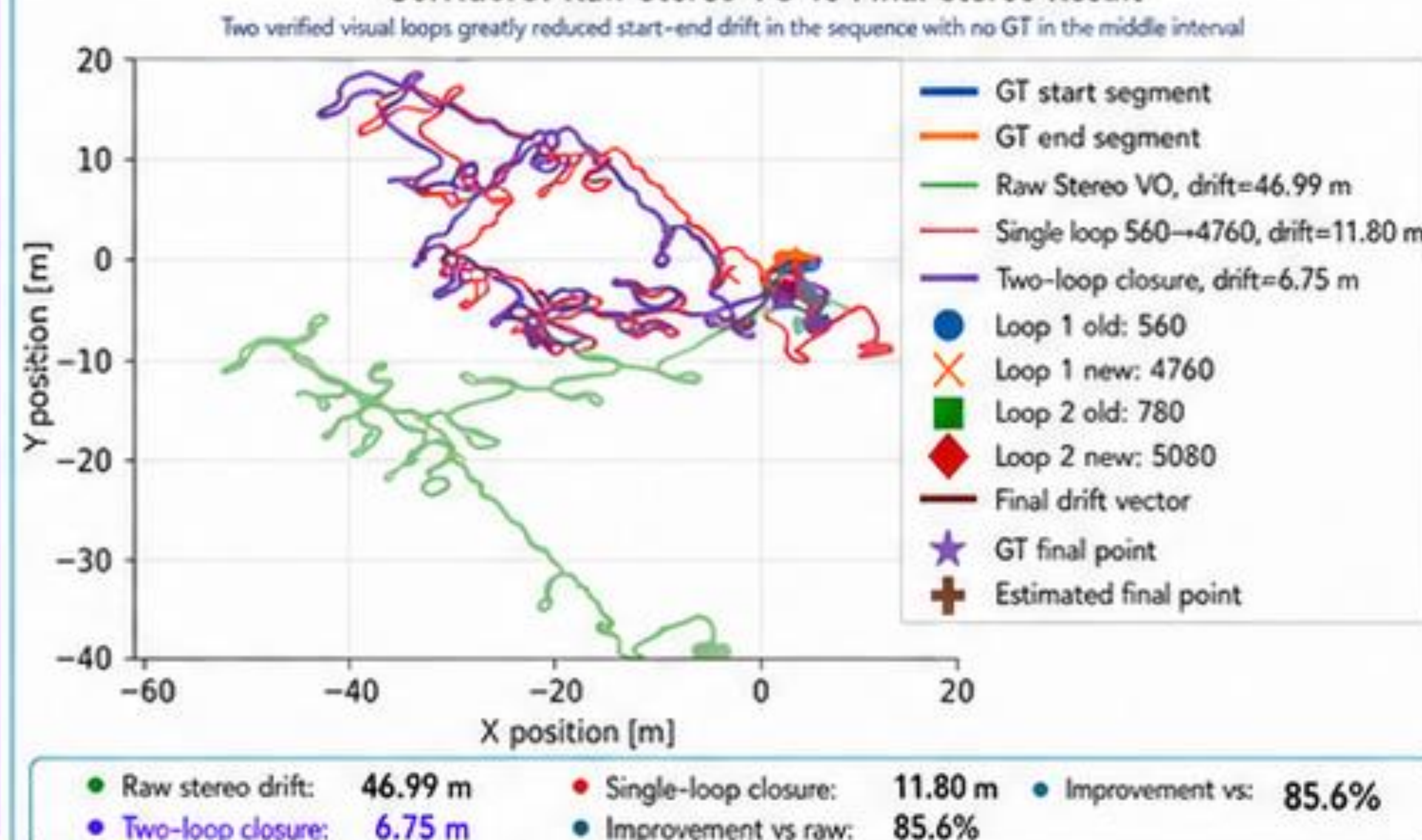


4. RESULTS

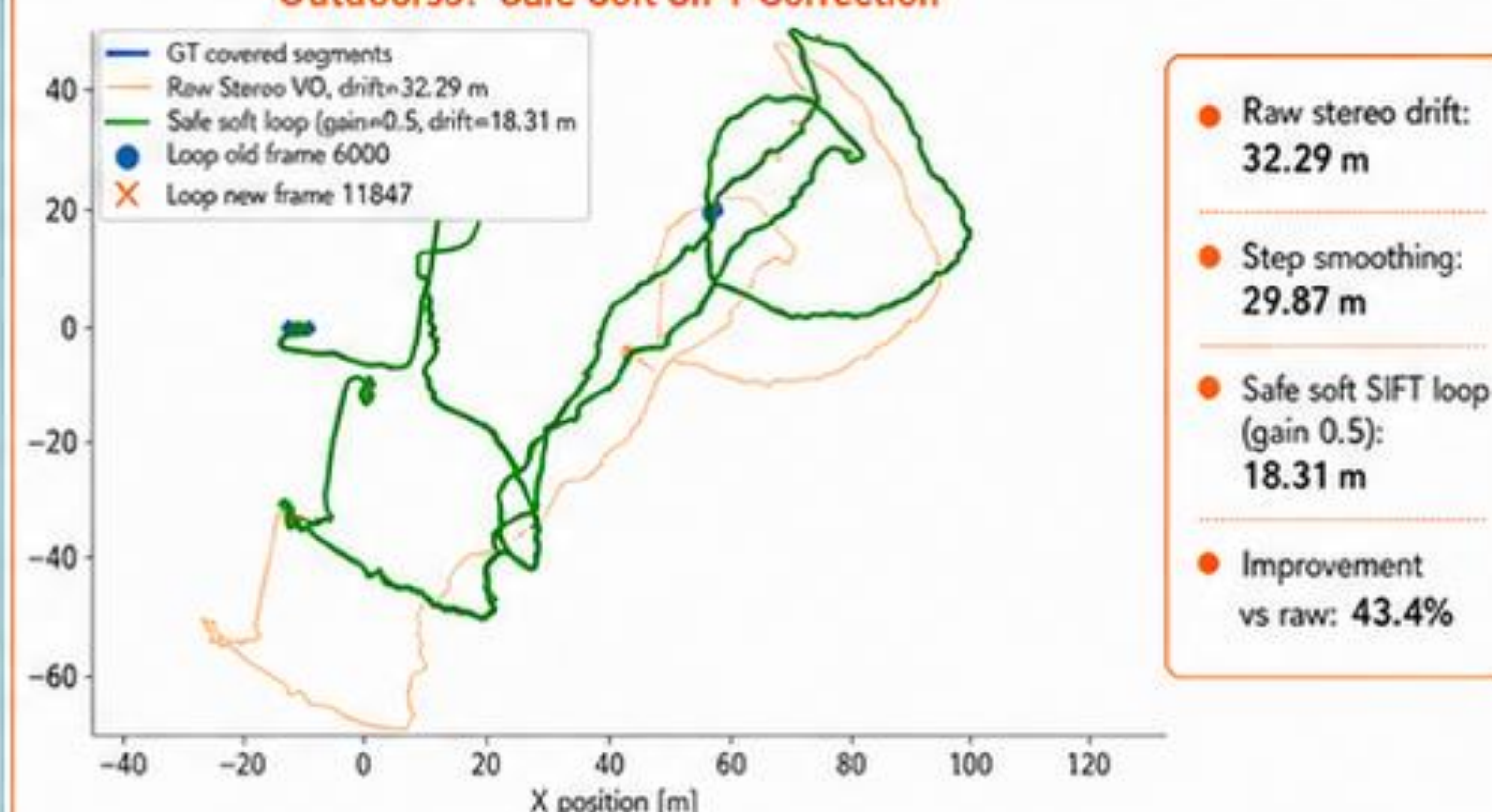
Room2: Monocular and Stereo VO vs Ground Truth



Corridor3: Raw Stereo VO vs Final Stereo Result



Outdoors5: Safe Soft SIFT Correction



5. PROBLEMS FACED & FIXES

Problems faced

- ✗ Fisheye rectification sensitivity
- ✗ Large accumulated drift in long trajectories
- ✗ Limited / sparse ground truth in Corridor3 and Outdoors5
- ✗ False or weak loop candidates in outdoor scenes
- ✗ Instability when loop correction is too strong

Fixes

- ✓ Used official TUM VI calibration and rectified stereo pairs
- ✓ Built robust ORB / SIFT matching with geometric verification
- ✓ Applied verified visual loop closure for Corridor3
- ✓ Used safe soft SIFT loop correction (gain = 0.5) for Outdoors5
- ✓ Kept GT for evaluation only, preserving the protocol

6. FINAL RESULTS SUMMARY

Dataset	Method	Main Metric	Result
Room2	Monocular VO	ATE / RPE	1.217 / 0.945 m
Room2	Stereo VO	ATE / RPE	0.944 / 0.339 m
Corridor3	Raw Stereo VO	Drift	46.99 m
Corridor3	Stereo + two loops	Drift	6.75 m
Outdoors5	Raw Stereo VO	Drift	32.29 m
Outdoors5	Stereo + safe soft loop	Drift	18.31 m

Runtime (approximate)

- Corridor3**
- Monocular VO: 154.96 s
 - Stereo VO: 284.05 s
- Outdoors5**
- Monocular VO: 604.98 s
 - Stereo VO: 1152.51 s

Hardware: Intel i7-1185G7, 32 GB RAM
(Results may vary on other systems)

7. CONCLUSION / TAKE-HOME MESSAGE

- Stereo VO provides metric motion estimation but can accumulate significant drift.
- Room2 shows the best stereo accuracy under full GT evaluation.
- Corridor3 demonstrates that verified visual loop closure can strongly reduce drift.
- Outdoors5 remains the most challenging sequence because of long trajectories, outdoor appearance changes, and false loop risk.

8. REFERENCES

- [1] Scaramuzza, D., and Fraundorfer, F. "Visual Odometry [Tutorial]." *IEEE Robotics & Automation Magazine*, 2011.
- [2] Schubert, D., Goll, T., Demmel, N., Usenko, V., Stuckler, J., and Cremers, D. "The TUM VI Benchmark for Evaluating Visual-Inertial Odometry." *IROS*, 2018.
- [3] Bradski, G. "The OpenCV Library." *Dr. Dobbs' Journal of Software Tools*, 2000.